



湖北大学曼城联合学院
Manchester Metropolitan Joint
Institute, Hubei University

ASSIGNMENT COVER SHEET

Unit code and title:	(6G5Z0042_2627_9FHU) - Full-Stack Web Development
Assignment set by:	Dr Xia Cui
Assignment ID:	1CWK100
Assignment weighting:	100%
Assignment title:	Implement the Auctionary API
Type: (Group/Individual)	Individual
Hand-in deadline:	Thursday 22th October 2026 – 2:00 pm (GMT Time)
Hand-in format and mechanism:	Your work must be submitted through Moodle. The deliverable is ONE zipped file containing: • All your code to run the website (front-end and back-end) • A screencast of the application

Learning Outcomes Being Assessed:

- | | |
|------------|---|
| LO1 | Implement high quality software to a predefined specification, applying a full-stack architecture using both front end and back-end web frameworks. |
| LO2 | Investigate and apply a series of existing tools, libraries, and techniques relevant to industry. |



Note: It is your responsibility to make sure that your work is complete and available for marking by the deadline. Make sure that you have followed the submission instructions carefully, and that your work is submitted in the correct format, using the correct hand-in mechanism (e.g., Moodle upload). If submitting via Moodle, you are advised to check your work after upload, to make sure it has uploaded properly. **Do not alter your work after the deadline.** You should make at least one full backup copy of your work.

The Timeliness of Submissions is Strictly Monitored and Enforced.

Joint institute students will follow Hubei's policies with respect to late submission. Where, following a late submission, a student has approved Exceptional Factors that cover the original submission period, the mark for the assessment will revert to the original, uncapped grade. Exceptional Factors shall be dealt with in accordance with Hubei University regulations. Students shall not be entitled to undertake reassessment in any assessment with a capped mark applied as a result of late submission.

Please note that individual tutors are unable to grant extensions to coursework. Extensions can only be granted on the basis of approved Exceptional Factors (see below).

Exceptional Factors affecting your performance: Students wishing to apply for Exceptional Factors should do so through the Hubei University process. The outcome of any application will be considered and decided by Hubei University and applied to both awards. If the application is accepted, the student shall have the right to be reassessed as if for the first time in any or all the elements of the assessment covered. Further details can be found in the [Joint Assessment Regulations](#).

Plagiarism: The Joint Institute students will follow Manchester Met policy for plagiarism. For further information on the Manchester Met policy, please see [Academic Integrity and Misconduct](#).

Plagiarism is the unacknowledged representation of another person's work, or use of their ideas, as one's own. Manchester Metropolitan University takes care to detect plagiarism, employs plagiarism detection software, and imposes severe penalties, as outlined in the Student Handbook. Bad referencing or submitting the wrong assignment may still be treated as plagiarism. If in doubt, seek advice from your tutor.

Use of generative AI: The use of generative AI is permitted in this assessment, so long as it is used in accordance with the instructions provided in the 'Are you allowed to use AI in assessments?' section of the [AI Literacy Rise Study Pack](#). All submitted work must be your own original content.

If you are unable to upload your work to Moodle: If you have problems submitting your work through Moodle, you can send your work to the Assessment Management Team using the [Contingency Submission Form](#). Assessment Management will then forward your work to the appropriate person for marking. If you use this submission method, your work must be



sent before the published deadline, or it will be logged as a late submission. Alternatively, you can save your work into a single zip folder, then upload the zip folder to your university OneDrive and submit a Word document to Moodle, which includes a link to the folder. **It is your responsibility to make sure you share the OneDrive folder with the Module Leader, or it will not be possible to mark your work.**

Your work will be submitted through Moodle, with Turnitin enabled. Turnitin is a tool used to identify plagiarism and collusion in written submissions.

Assessment Criteria:	As indicated in the attached assignment specification
Formative Feedback:	Lecture/Lab discussion and interactions with tutor onwards from when the assignment is set; You will also be provided with a set of test scripts for assessing your progress on part of the assignment.
Summative Feedback Format:	You will be given individual feedback via Moodle; Common feedback for the entire class will also be posted on Moodle.



Full Stack Web Development Assignment

– Implement the Auctionary API

Introduction

This assessment is coursework based, and worth 100% of the overall unit mark. The tasks that you are required to complete for this assessment are outlined in this coursework specification.

Aim

This unit encourages you to analyse real world situations critically. The assessment mimics industry projects by requiring you to engage with multiple disciplines. By the end of the unit, you will have completed the development of a full-stack application that uses a variety of advanced web technologies. It is encouraged that you maintain a portfolio of projects throughout university (e.g., through GitHub) that can serve as a portfolio of your work when applying for jobs and placements. This project could serve as one aspect of your portfolio.

The following skills will be essential for successful completion of this coursework (and including such a project in your portfolio would demonstrate these skills to potential employers):

- Real world problem solving: You will need to analyse a real-world situation, develop solutions for multiple problems when developing the application, and then evaluate your solutions.
- Technical skills: This assessment requires you to write an application in JavaScript using NodeJS, the ExpressJS framework, and VueJS for the front end. In addition to these technologies, you will gain an understanding of RESTful APIs and the OpenAPI specification.
- Understanding of modern relevant JavaScript frameworks: NodeJS, ExpressJS and VueJS are all highly utilised frameworks in industry, and engineers with an understanding of such frameworks are highly sought after.

Assessment Learning Outcomes

LO1: Implement high quality software to a predefined specification, applying a full-stack architecture using both front end and back-end web frameworks.

LO2: Investigate and apply a series of existing tools, libraries, and techniques relevant to industry.

Coursework Overview

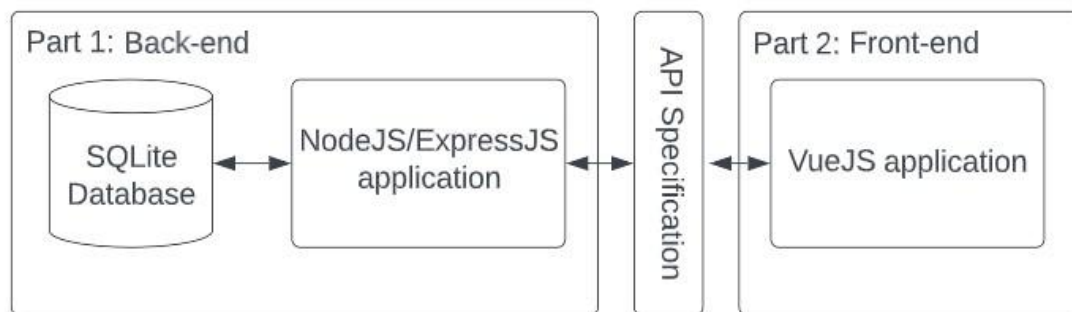
To complete this assessment, you are required to develop a full-stack web application. The precise detail of the coursework tasks is detailed in section four below. However, to summarise, the assignment is split into two parts. You will be provided with an API specification for a RESTful API. In the first part of the assignment, you will develop the back-end application that implements the API specification. In the second part of the assignment, you will develop the front-end application that interacts with the API to provide functionality to its users. You are required to write both applications using the frameworks and technologies taught throughout the Unit in the lectures and the labs.

The Assessment (1CWK100)

Scenario

You have been hired to implement a prototype of a new product, an auction management tool called “Auctionary”.

You’ll be familiar with similar applications. However, their monolithic structure has made them very slow and clunky as they have grown. This has led to many users migrating to other competitor platforms and has had a huge effect on profits. You will write the platform using modern frameworks, completely splitting the back-end from the front-end so that both can scale independently of each other. A senior solutions architect has already developed an API specification and database structure to get you started, and a quality assurance team have also drawn up a series of test scripts based on the API specification so that you can ensure that your application conforms to the API specification.



Detailed Specification

Part 1: Implement the back-end

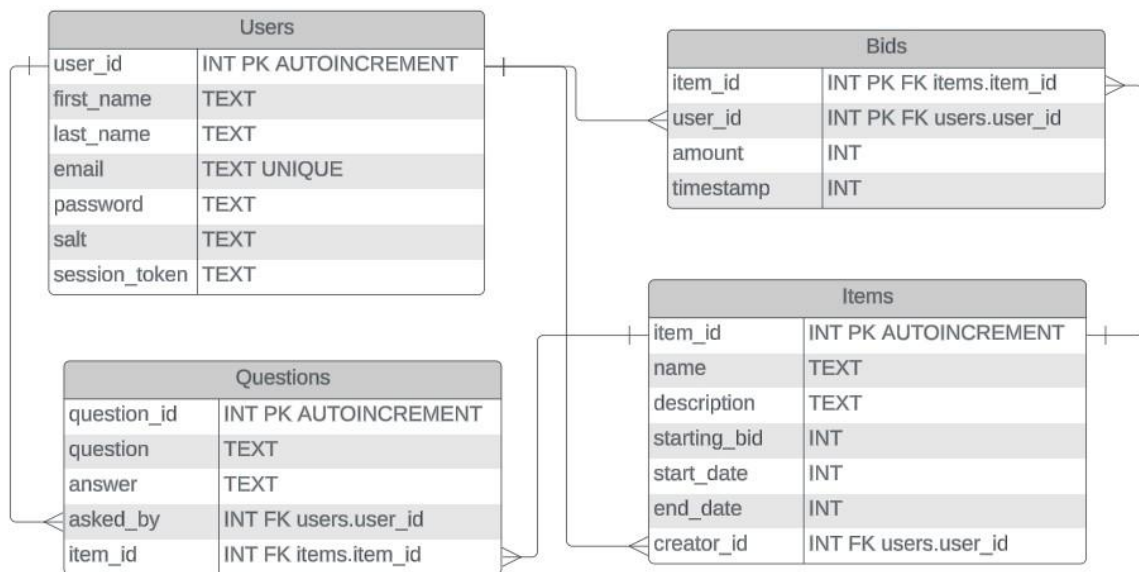
The API specification and database structure have been provided for you. **It is important that you adhere to the exact specification as you are being marked on your ability to implement it (as would be expected in industry).** A set of test scripts have also been provided for you to ensure that your application meets the specification.

The API specification

The API consists of a series of end-points for you to implement. The labs and lectures will walk you through implementing the first couple of end-points and then you will complete the rest for the assignment. The latest version of the API specification is available to see on Swagger (we will talk about Swagger during week 1): <https://app.swaggerhub.com/apis/XCUI/Auctionary/1.0.0>

The database structure

When you run the starter code (details below), a SQLite database will be created. SQLite is a small SQL database that runs on a single file (you will see a file named db.sqlite in your project directory). Running the starter code will also automatically create the tables necessary for the assignment. The tables will be structured as outlined below:



The starter code

A starter project has been provided for you. This is to provide you with the database and test scripts. An explanation of the project structure, along with instructions on downloading and running the starter project will be provided in the lectures and labs. All your code should be written inside the /app/ directory, and you shouldn't edit any of the other existing files unless instructed to do so.

The starter code is available here: https://gitcode.com/MMIJE/auctionary_starter

Running the tests

As mentioned, the starter code includes a series of test scripts for checking that your code adheres to the API specification. These scripts are provided to you for checking and will also be used to mark a portion of the assignment. To run the tests, you will need to have two terminal windows open: in the first you will need to run your server (`npm run dev`), and in the second you can run your tests (`npm test`). I will demonstrate this in more detail in the lectures and labs.

Note: When marking, different data will be used and additional tests ran to ensure that your code hasn't been tailored to only respond to certain data. Your code will also be manually reviewed to ensure proper implementation.

Part 2: Implement the front-end

Part two of the assignment is to create the front-end interface for the users and audience of the application. This front-end should interact with the API to provide functionality and be created using the VueJS framework. Instructions on how to get set up with VueJS will be provided in the lectures and labs.

Auctionary is intended to be a generic template for creating auction sites. We sometimes refer to this as a "white-label product" (think about how WordPress gives us a generic template for creating blogs, but those blogs can be about anything). You should showcase your frontend by applying a specialty brand that is relevant to your interests – this could be anything (e.g. Pokemon cards, cars, golf clubs, vinyl, whiskey). The brand of your application should fit to the specialty you choose (e.g. Pokemon cards may look fun with bright colours whereas whiskey would be more high-end and crisp).



Getting Started and Recommended Order

You will be assessed based on your applications coverage of the entire API (both back-end, frontend, and extension tasks). The mark scheme is detailed in the marking rubric at the end of this document.

Each week you have been provided with a checklist of activities for working towards your assignment submission. The lectures and labs will provide guidance on how to complete the checklist items for each particular week. The order in which you complete tasks is up to you. However, it is recommended that you follow along with the weekly checklist to ensure that nothing is missed.

The checklist and weekly breakdown can be found below.

Week-Session	Lecture	Lab	Assignment Checklist	
1-1	Introduction to Full-Stack	RESTful APIs	☐	Familiarise yourself with RESTful API concepts
1-2	Introduction to NodeJS/Express	Intro to NodeJS/Express	☐ ☐	Familiarise yourself fully with the assignment API Download and run the starter code project
2-1	Interacting with Databases	Interacting with Databases	☐	Using the lab to help you, implement the user management end-points
2-2	Handling Authentication	Implementing Authentication	☐	Using the lab to help you, implement the user management end-points
3-1	Version Control	Assignment Support: Backend	☐ ☐	Complete all end-points Ensure your application passes all tests
3-2	API Security and Scaling	Assignment Support: Backend	☐ ☐	Implement version control for your assignment Implement extension task 1 and 2 (detailed below)
4-1	Introduction to VueJS	Intro to VueJS	☐	Create a new VueJS app for your assignment
4-2	Style, Layout and Navigation	Layout and Navigation	☐	Using the lab, design your application and create each of the pages and forms necessary
5-1	Networking	Handling API requests	☐	Using the lab, begin interacting with the API to provide functionality to your app
5-2	Structuring and Refactoring	Assignment Support: Front-end	☐	Complete all front-end functionality



6-1	Code Quality	Assignment Support: Front-end	☐	Implement extension task 3 (detailed below)
6-2	Assignment Support	Assignment Support	☐	Prepare assignment submission



Extension Tasks

Extension task 1 – Back-end (easy difficulty): Create a profanity filter on new auction items and questions to ensure any offensive language does not make it into the application (perhaps you can find a library to help?).

Extension task 2 – Back-end (hard difficulty): Design and implement a categories system for the application. This will require a DB table of categories, an endpoint for retrieving the categories, and then edits to other end-points and DB tables to store the category info for each event. The search endpoint should then also implement a filter for searching by category. Make this more difficult by allowing items to be associated with multiple categories.

This is only to be attempted after you have completed both applications. Your code needs to run without error and the tests need to be passing for the marks.

Extension task 3 – Front-End (hard difficulty): Alter your front-end application so that users can save local drafts of items before sending them to the API. You will need to save these drafts to the local storage within the browser and have functionality to view, edit and delete these drafts (much in the same way that mail clients work).

What do you need to get working to pass?

This assignment a substantial solo project, and sometimes we find that students can be overwhelmed with where to start. Therefore, we recommend you follow along with this section first, and then get your work checked over in the labs to ensure that you're meeting the specification.

The way the module is taught means that you might make more progress on the backend before moving onto the front-end elements. That's absolutely fine – this is just here to serve as a bare minimum application that you need to get working to get to 40%.

To get 40%, you must have:

1. All user management endpoints fully working
2. A front end that successfully implements the user management end points

All the code for authentication is provided, so it should be a case of following along with the lab sheets.

Support, Feedback, and Resources

As you can see from the checklist table above, the weekly assignment activities map nicely onto the lectures and lab content for each week. If you are struggling with a particular concept or task, the first place to check should be the relevant week's teaching material. If you require support or have queries that you can't find an answer to in the teaching material, you should ask your tutor in your timetabled lab. They will help you or point you to any relevant resources necessary.

With the focus of the lab sessions being on the assignment application, you can also use the lab sessions to get feedback and advice from the tutor with regards to your assignment.

Please don't share any assignment code with other students unless it has been provided to you in the labs/lectures. Sharing code could lead to you being reported for plagiarism.

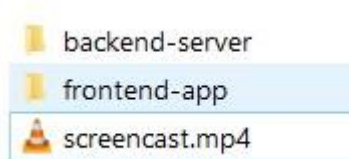


Submission Instructions

Submission of this coursework will be online, through the university's Virtual Learning Environment (Moodle). You must upload **a single zip file**, which includes the following:

1. A directory containing all your source code for the back-end server (delete your node_modules directory before submitting)
2. A directory containing all your source code for the front-end application (delete your node_modules directory before submitting)
3. A screencast of approximately 5 minutes, in mp4 format. In the screencast, you should walk through all the front-end applications' functionality, along with any other features you wish to highlight (OBS Studio is a free screen recorder that is easy to use <https://obsproject.com/>)

The structure of your zip file should look like the below:



Note: Moodle has a 100MB limit on uploads. Following the instructions above (deleting any node_modules directories and using MP4 format on the screencast) should mean that your zip file easily fits under that limit. However, if your submission is still creeping over 100MB:

1. Upload your zip file to your University OneDrive account
2. Create a share link to your directory (make sure it is shared publicly)
3. Upload a zip file to Moodle containing a txt file with the share link

The University has a strict policy on late submissions. It is your responsibility to ensure that your assignment is uploaded on time. Remember, large files will take time to upload.



Assessment Marking Criteria							
	Fail (0 to 29%)	Marginal Fail (30 to 39%)	3 rd Class (40 to 49%)	2 nd Class: 2 (50 to 59%)	2 nd Class: 1 (60 to 69%)	1 st Class (70 to 85%)	Exceptional 1 st (86 to 100%)
Back-end server <i>(assessed via automated tests and source code)</i> 45%	The backend functionality will be assessed through a collection of automated tests. These tests will assess your application against the API specification, including all input validation and response codes. Source code will also be assessed, and marks will be adjusted accordingly for implementations that differ from what has been taught in the labs (e.g., not using the database provided). Note: - Not all tests are equal, and some are repeated. You are being assessed on the proportion of API covered, not the number of tests you are passing. - These automated scripts also check for plagiarism and similarity across all other submissions.						
Front-end application: Functionality <i>(assessed via source code and screencast)</i> 30%	Marks for frontend functionality will be provided as a percentage of the functionality completed (e.g., if you complete half of the end-points, you will get 50%). Partial marks will be awarded for functionality which has been attempted.						
Front-end application: User Experience <i>(assessed via source code and screencast)</i> 15%	Very little consideration to usability. No branding attempted.	A demonstrated basic understanding of usability concepts.	A natural feel to app navigation and usability. Some common theme and branding are present, but there are inconsistencies throughout.	Consistent style, navigation, and usability features throughout the app. Some error handling.	Use of a style framework for handling usability. Natural feel to navigation and consistent throughout. Good error handling.	Excellent use of style and usability, with some consideration to accessibility. Handles all errors gracefully with validation and checking.	Exceptional style, usability, use of frameworks and accessibility features. Exceptional branding, relevant to the brand portrayed.
Extension Tasks <i>(assessed via source code and screencast)</i> 10%	No extension tasks attempted.	A demonstrated partially working solution to at least one of the extension tasks.	At least one of the extension tasks works as expected.	At least two of the extension tasks works, albeit with a few minor bugs or inefficiencies.	All extension tasks work, albeit with a few minor bugs or inefficiencies.	Two tasks are completed to a professional, fully functioning standard. The other is mostly complete.	All tasks fully complete to a professional standard.